

Andrew McElhannon

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EDUCATION

Shiley-Marcos School of Engineering, University of San Diego

B.A. Computer Science

Final G.P.A: 3.9/4.0

San Diego, CA

May 2026

RELEVANT COURSEWORK

Principles of Digital Hardware; Operating Systems; Networking; Database Management Systems Design; Object Oriented Programming; Intro to Cyber Concepts and Tools, Cybersecurity Network Defense

SKILLS/ CERTIFICATIONS

Languages: Python, Java, C, SQL, HTML/CSS, SystemVerilog

Tools/Technologies: Git, GitHub, Linux, REST APIs, Socket Programming, Virtual Machines, pfSense, Nessus, iOS, Windows

Certifications: Security+, Network+

UNIVERSITY PROJECTS

Adversys (Senior Project), University of San Diego

Fall 2025 - Spring 2026

- **Collaborating** with a four-person development team to enhance an existing cybersecurity application and support deeper integration of embedded AI agents and security tools
- **Modifying** the existing codebase to enable AI agents to identify vulnerabilities using Adversys security tooling and support automated remediation-report generation
- **Designing** the integration with an emphasis on safe, secure, and reliable automation of vulnerability assessment and remediation **workflows**
- Developing within an **Agile workflow** using two-week sprints, peer reviews, technical presentations, and **iterative software development**

Streaming Music Protocol, University of San Diego

Spring 2026

- **Designed** a client-server music streaming application using TCP sockets to transmit binary audio and resource requests
- Implemented a custom **application-layer protocol** and server-side parser supporting request validation and error detection
- Resolved cross-language binary compatibility between Java and C, including byte ordering, **memory alignment**, and **data serialization**

SurfBuddy, University of San Diego

December 2024

- Co-engineered a **scalable** terminal application integrating **APIs** to fetch, parse, and display real-time environmental data
- Applied **object-oriented** design and the Factory Method pattern to construct location and preference objects while separating object creation from application logic

Cybersecurity Network, University of San Diego

Fall 2025

- Co-Architected a **network of multiple Linux-based VMs** on a physical host to simulate attacks and network routing/management
- Configured and deployed a **pfSense** firewall as a default gateway to audit and protect network traffic
- Executed vulnerability assessments using Nessus scans and performed brute-force password attacks

Generallt, Jean Cabot University

April 2025

- **Designed** and deployed a **minimum viable prototype** e-commerce website using HTML/CSS
- Engineered a responsive frontend layout with intuitive **UI**, **clear navigation**, and a modern aesthetic design
- Demonstrated **intended functionality** and user flows that could be shown to a potential client